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ORIGINAL ARTICLE

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Topological superconductivity in multifold fermion metals

Zhe Shen Gao^{1†} , Xue-Jian Gao^{1†}, Wen-Yu He², Xiao Yan Xu³, T.K. Ng¹ and K.T. Law^{1*}

Abstract

Recently, multifold fermions characterized by band crossings with multifold degeneracy and Fermi surfaces with higher Chern numbers have been discovered experimentally in AlPt (Schroter et al. in *Nat Phys* 15:759–765, 2019) and XSi (X = Rh, Co) (Sanchez et al. in *Nature* 567:500–505, 2019; Rao et al. in *Nature* 567:496–499, 2019; Takane et al. in *Phys Rev Lett* 122:076402, 2019). In this work, we largely expand the family of multifold fermion materials by pointing out that several well-studied noncentrosymmetric superconductors are indeed multifold fermion metals. Importantly, their normal state topological properties, which have been ignored in previous studies, play an important role in the superconducting properties. Taking $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ as examples, we found a large number of unconventional degenerate points, such as double spin-1, spin-3/2, Weyl and double Weyl topological band crossing points near the Fermi energy, which result in finite Chern numbers on Fermi surfaces. Long Fermi arc states in $\text{Li}_2\text{Pd}_3\text{B}$, originating from the nontrivial band topology were found. Importantly, it has been shown experimentally that $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ are fully gapped and gapless superconductors, respectively. By analyzing the possible pairing symmetries, we suggest that $\text{Li}_2\text{Pd}_3\text{B}$ is a conventional s-wave superconductor or DIII class topological superconductor with Majorana surface states. $\text{Li}_2\text{Pt}_3\text{B}$, being gapless, is likely to be a nodal topological superconductor with dispersionless surface Majorana modes. We further identified that several noncentrosymmetric superconductors, such as $\text{Mo}_3\text{Al}_2\text{C}$, PdBiSe , Y_2C_3 and La_2C_3 , are multifold fermion superconductors. This work calls for a revisit for the study of noncentrosymmetric superconductors which provide platforms for investigating the interplay between superconductivity and topologically nontrivial Fermi surfaces.

Keywords: $\text{Li}_2(\text{Pd}, \text{Pt})_3\text{B}$, Multifold fermion, Topological superconductivity

1 Introduction

Noncentrosymmetric superconductors are superconductors without inversion centers. The broken inversion symmetry results in antisymmetric spin-orbit coupling (ASOC) which can lead to interesting superconducting properties, such as mixed pairing order parameters [5–7], helical phases [8–10], novel magnetoelectric responses [11–14], large enhancement of the upper critical field H_{c2} [14, 15] or topological superconductivity [16–20]. In the

past two decades, many noncentrosymmetric superconductors have been discovered [21]. To understand the superconducting properties of noncentrosymmetric superconductors, a common procedure was to construct normal state Hamiltonians which respect the crystal symmetry, and then to further include ASOC and study its effect on superconductivity. However, the normal state topological properties of these noncentrosymmetric superconductors have usually been ignored in previous studies.

In recent years, tremendous progress had been made in the understanding of the topological properties of band structures. Particularly, many topologically nontrivial band crossings which describe unconventional fermions, such as Weyl, Kramers Weyl and unconventional multifold fermions have been discovered [1–4, 22–25].

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These multifold band crossings give rise to finite Chern numbers on Fermi surfaces enclosing the band crossing points. The case of multifold fermions is particularly interesting, as the nonsymmorphic and time-reversal (TR) symmetry can enforce multiple degeneracies and result in large Chern numbers on Fermi surfaces enclosing the band crossing points, even in the absence of ASOC. Unlike Weyl points, which usually result in short Fermi arc states, multifold fermion crossing points result in incredibly long Fermi arc states which span a large portion of the surface Brillouin zone. So far, two multifold fermion semimetals, namely, AlPt [1] and XSi (X = Rh, Co) [2–4], have been identified, and their long Fermi arc states have been observed through ARPES experiments recently. However, they are not superconducting, and it is not known how the normal state topology resulting from multifold fermions can affect the superconducting properties in realistic materials.

In this work, we point out that $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$, which are well-studied noncentrosymmetric superconductors [26–29], are superconducting multifold fermion metals and the normal state topological properties of these materials have been ignored in previous studies. Through NMR [28–30], specific heat [31, 32] and penetration length [27] measurements, it has been suggested that $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ are fully gapped and gapless superconductors, respectively. We first identify topological band crossing points in $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ which are relevant to the Fermi surface topology through symmetry analysis and *ab initio* calculations. Several 2-fold, 4-fold and 6-fold band crossings which correspond to Weyl, Kramers Weyl, spin-1 and multifold fermions were found and are listed in Table 1. Fermi arcs which span long distances in the surface Brillouin zone on the (001) surfaces are found in $\text{Li}_2\text{Pd}_3\text{B}$. Second, by introducing superconducting pairing to the bands near the Fermi energy, we show that the gapless superconducting states of the A_1 and A_2 representations, which are relevant to $\text{Li}_2\text{Pt}_3\text{B}$, are topological and possess dispersionless Majorana surface modes. A fully gapped DIII topological superconducting phase in $\text{Li}_2\text{Pd}_3\text{B}$ is also discussed. Finally, we point out that several other well studied noncentrosymmetric superconductors, such as $\text{Mo}_3\text{Al}_2\text{C}$, PdBiSe , Y_2C_3 and La_2C_3 are also multifold fermion metals with rich topological band structures. This

Table 1 Unconventional fermions found in $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ by first-principles calculation. Indices correspond to those in Fig. 1

Type	Monopole charge	Degeneracy	No.
Weyl	± 1	2	2, 3
Kramers Weyl	± 1	2	6, 7, 13
Double Kramers Weyl	± 2	4	1, 4, 10, 11
Double Spin-1	± 4	6	9, 15
Spin-3/2	± 4	4	5, 8, 12, 14

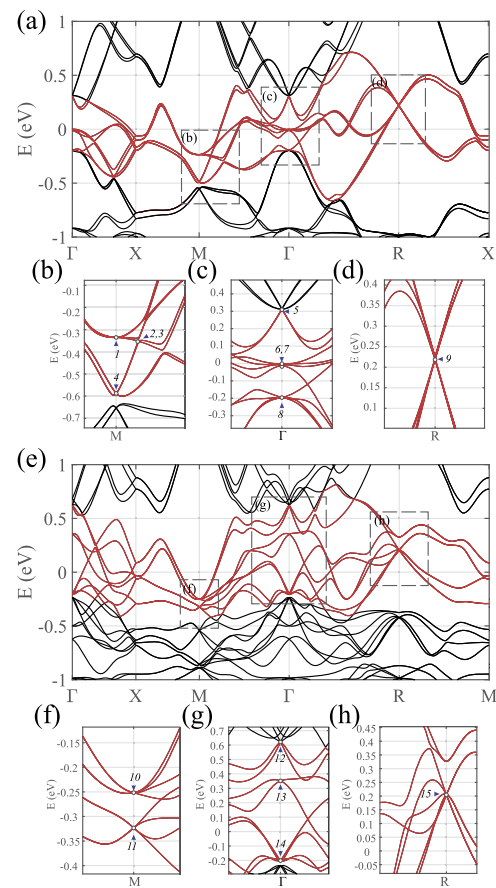


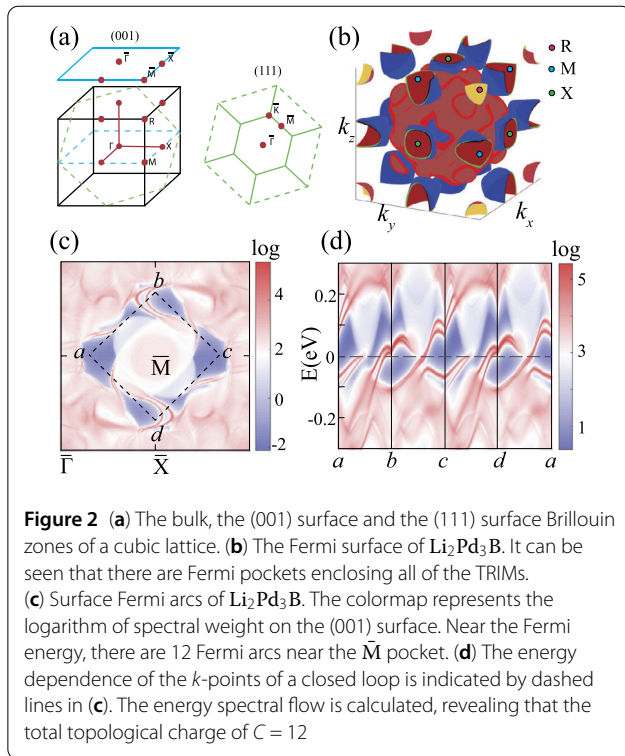
Figure 1 Band structure of $\text{Li}_2\text{Pd}_3\text{B}$ (a-d) and $\text{Li}_2\text{Pt}_3\text{B}$ (e-h) from first-principles calculations with SOC self-consistently included. Bands that cross the Fermi energy are marked in red, and the topological band crossing points which can be described by unconventional fermions are depicted in enlarged plots. The degeneracies corresponding to unconventional fermions are enumerated and listed in Table 1

work motivates a revisit for the study of noncentrosymmetric superconductors which are platforms for investigating the interplay between superconductivity and topologically nontrivial Fermi surfaces.

2 Multifold fermions in $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$

Both $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ have a cubic structure, and they belong to the chiral space group $P4_332$ (No. 212) with point group O [33]. Its generators contain the following (screw) rotation operations (in Seitz symbol):

$$\left\{ 2_{001} \left\| \frac{1}{2}, 0, \frac{1}{2} \right\|, \quad \left\{ 2_{010} \left\| 0, \frac{1}{2}, \frac{1}{2} \right\|, \right. \right. \\ \left. \left\{ 3_{111}^+ \parallel 0 \right\}, \quad \left\{ 2_{110} \left\| \frac{1}{4}, \frac{3}{4}, \frac{1}{4} \right\| \right\}. \quad (1)$$



The chiral multifold fermions originate from these screw rotations [22]. *Ab initio* calculations (see Additional file 1) are used to obtain the band structures of $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$, as depicted in Fig. 1. In reciprocal space, the cubic Brillouin zone has nonequivalent high symmetry points $\Gamma(0, 0, 0)$, $X(\pi/a, 0, 0)$, $M(\pi/a, \pi/a, 0)$ and $R(\pi/a, \pi/a, \pi/a)$ (with primitive lattice constant a), as shown in Fig. 2a. The band structures of $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ were calculated more than a decade ago [33], but the unconventional crossing points that correspond to the multifold fermions have not been identified. Recently, it was pointed out in [22] that $\text{Li}_2\text{Pd}_3\text{B}$ possesses a six-fold degenerate point at the R point. In this work, we point that there are indeed a large number of unconventional crossing points near the Fermi energy of $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ and these topological crossing points have important impacts on the topological properties of the Fermi surfaces and the superconducting states.

In $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$, time-reversal symmetry forces two copies of spin-1 fermions to be degenerate at R [34], giving a 6-fold degeneracy as depicted in Fig. 1d. This is referred to as double spin-1 in Table 1. On the other hand, we found a 4-fold degeneracy at Γ (e.g., labeled as No. 14 in Fig. 1g) corresponds to the spin-3/2 or the so-called Rarita-Schwinger-Weyl (RSW) fermions with a monopole charge of ± 4 . The corresponding irreducible representations and an effective $k \cdot p$ Hamiltonian linear in k for the two aforementioned unconventional fermions are included in the Additional file 1. For the bands which cross the Fermi en-

ergy (denoted in red in Fig. 1), 15 unconventional degenerate points are identified which are associated with unconventional chiral fermions. The locations of these points are labeled in Fig. 1, and the degeneracy and the topological charges of the Fermi surfaces enclosing these points are summarized in Table 1.

Focusing on the $\text{Li}_2\text{Pd}_3\text{B}$ case, the Kramers Weyl fermions at M , which is a TRIM, are doubled by non-symmorphic symmetries, giving double Kramers Weyl fermions [34]. We further identified two Weyl fermions (No. 2, 3 in Table 1) along the M - Γ line with a total Berry curvature charge of +2. It is interesting that these Weyl points have to be taken into account for the observed surface spectrum.

When a nondegenerate Fermi surface in the Brillouin zone encloses the band crossing points, it can acquire a finite Chern number, and there can be surface Fermi arc states connecting surfaces with different Chern numbers. The Fermi surface of $\text{Li}_2\text{Pd}_3\text{B}$ is depicted in Fig. 2b. Figure 2c visualizes the calculated surface spectral function $\mathcal{A}_s(E, \mathbf{k}) = -\frac{1}{\pi} \text{Im Tr } G_s(E, \mathbf{k})$ of the (001) surface of $\text{Li}_2\text{Pd}_3\text{B}$, where $G_s(E, \mathbf{k})$ is the surface Green's function. As many as 12 Fermi arcs emerge from the $\bar{M}$ pocket. The $\bar{M}$ surface pocket is a projection of the R and M pockets of the 3D Brillouin zone, and the rest of the states enclosing the $\bar{M}$ pocket are projections of pockets around the Γ point in the 3D Brillouin zone. The energy dispersion of the Fermi arc states are depicted in Fig. 2d. Note that all the Fermi arc states around the $\bar{M}$ pocket have positive Fermi velocity along the a - b - c - d lines defined in Fig. 2c. This suggests that the gapped surface enclosing the M and R points in the 3D Brillouin zone has a total Chern number of 12. This is consistent with the fact that the two Fermi surfaces at the R point carry a total monopole charge of 4 and the eight Weyl points (No. 2, 3 in Table 1 together with Weyl points related by 4-fold rotation symmetry) near M carry a total monopole charge of 8.

For the case of $\text{Li}_2\text{Pt}_3\text{B}$, the effect of ASOC is stronger than that in $\text{Li}_2\text{Pd}_3\text{B}$ which can be seen from the larger band splitting. Larger ASOC might affect superconductivity as will be discussed later, but topology are essentially the same. Band dispersions are more complicated for $\text{Li}_2\text{Pt}_3\text{B}$, making it impossible to separate Fermi arcs from bulk states in surface projection.

3 Gapless pairing in $\text{Li}_2\text{Pt}_3\text{B}$

As shown in the previous sections, the band structure of $\text{Li}_2\text{Pt}_3\text{B}$ is highly topologically nontrivial, and the Fermi surface topological properties cannot be captured by simple effective models. All the nontrivial crossing points enclosed by the Fermi surface has to be taken into account, which determines the Chern number of the Fermi surface. Therefore, in our calculations, we first construct the Wannier orbitals from first-principles calculations with

Table 2 Possible pairing phases of $\text{Li}_2\text{Pt}_3\text{B}$ labeled by the 1D IR of O point group. In each IR, the pairing function $\Delta_{\nu,\mathbf{k}}$ can be expanded in terms of the basis functions. The η_ζ with $\zeta = 1, 2, 3, \dots$ there are the coefficients for the basis functions in A_1 and A_2 IR respectively

IR	Pairing function $\Delta_{\nu,\mathbf{k}}$
A_1	$\eta_1 + \eta_2(3 - \cos k_x a - \cos k_y a - \cos k_z a) + \dots$
A_2	$\eta_3(\cos k_x a - \cos k_y a)(\cos k_y a - \cos k_z a) \times (\cos k_z a - \cos k_x a) + \dots$

the *Wannier90* package [35] and the Hamiltonian is diagonalized to obtain the eigenstates $|\phi_{\nu,\mathbf{k}}\rangle$. The realistic band structures obtained allows us to fully incorporate the topological properties of the normal state Fermi surface when studying the superconducting phase. For simplicity, we assign the same pairing order parameter $\Delta_{\nu,\mathbf{k}}$ to all the bands. The pairing $\Delta_{\nu,\mathbf{k}}$ transforms as one of the irreducible representations (IR) of the point group O, which is the factor group of $P4_332$. With this method, we can classify all the symmetry-allowed pairing functions up to their leading order in k corresponding to the irreducible representations of point group O, as listed in the Table S1 of Additional file 1.

Using Ginzburg-Landau analysis, higher dimensional representations either break crystal symmetries or time-reversal symmetry [36–39] (see also Additional file 1), which has not been observed experimentally. Therefore, in this work, we focus on pairings associated with the one-dimensional representations. The space group $P4_332$ allows two one-dimensional representations, A_1 and A_2 , as listed in Table 2. The pairing in the A_1 representation includes the conventional s-wave and the extended s-wave pairing and the pairing in the A_2 representation is gapless, as it requires a sign change of order parameter under C_4 rotation.

$\text{Li}_2\text{Pt}_3\text{B}$ has been experimentally found to be a gapless superconductor through penetration length [27], NMR [29, 30] and specific heat measurement [31, 32]. This was suggested to be a consequence of ASOC in the previous literature [27]. From the symmetry analysis, we find that the A_2 pairing phase with $\Delta_{\nu,\mathbf{k}} = \eta_3(\cos k_x a - \cos k_y a)(\cos k_y a - \cos k_z a)(\cos k_z a - \cos k_x a)$ satisfies the gapless condition. It is evident from the pairing function that there are many nodal planes such as the $k_x = k_z$ plane. When the nodal planes intersect with the Fermi surface, nodal lines are formed, e.g., bright lines in Fig. 3a. The projection of the nodal lines on the surface Brillouin zone can create pockets which are filled by dispersionless Majorana modes on the (111) surfaces, as shown in Fig. 3b. To understand the topological origin of the dispersionless Majorana modes on the (111) surface, we can parameterize the Hamiltonian as $\mathcal{H}_{\text{BdG}}(\bar{k}_x, \bar{k}_y, \bar{k}_z)$, where $\bar{k}_z$ is along the (111) direction and $\bar{k}_x$ and $\bar{k}_y$ are momenta parallel to the (111) surface. By taking $\bar{k}_x$ and $\bar{k}_y$ as numbers, $\mathcal{H}_{\text{BdG}}(\bar{k}_x, \bar{k}_y, \bar{k}_z)$ can

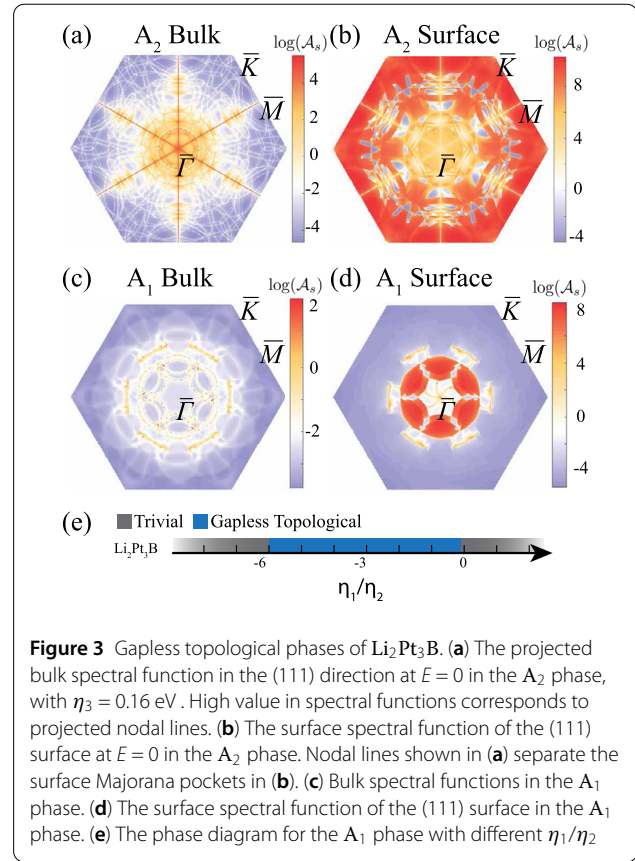


Figure 3 Gapless topological phases of $\text{Li}_2\text{Pt}_3\text{B}$. (a) The projected bulk spectral function in the (111) direction at $E = 0$ in the A_2 phase, with $\eta_3 = 0.16$ eV. High value in spectral functions corresponds to projected nodal lines. (b) The surface spectral function of the (111) surface at $E = 0$ in the A_2 phase. Nodal lines shown in (a) separate the surface Majorana pockets in (b). (c) Bulk spectral functions in the A_1 phase. (d) The surface spectral function of the (111) surface in the A_1 phase. (e) The phase diagram for the A_1 phase with different η_1/η_2

be regarded as a 1D AIII class Hamiltonian $\mathcal{H}_{\text{BdG},\bar{k}_x,\bar{k}_y}(\bar{k}_z)$ which depends on $\bar{k}_z$ only. The topological invariant can be written as:

$$N_{\bar{k}_x,\bar{k}_y} = \frac{1}{2\pi} \text{Im} \int d\bar{k}_z \text{Tr}[\partial_{\bar{k}_z} \ln Q(\bar{k}_x, \bar{k}_y, \bar{k}_z)], \quad (2)$$

which determines the number of zero energy Majorana modes at the point $(\bar{k}_x, \bar{k}_y)$ of the surface Brillouin zone [40–42]. Here, $Q(\bar{k}_x, \bar{k}_y, \bar{k}_z) = H_0(\bar{k}_x, \bar{k}_y, \bar{k}_z) - i\Delta(\bar{k}_x, \bar{k}_y, \bar{k}_z)$ and $H_0(\bar{k}_x, \bar{k}_y, \bar{k}_z)$ is the normal state Hamiltonian.

It is important to note that the A_1 pairing phase can also result in a gapless spectrum. As shown in Fig. 3e, there is a wide range of parameter regime of η_1/η_2 for A_1 pairing $\Delta_{\nu,\mathbf{k}} = \eta_1 + \eta_2(3 - \cos k_x a - \cos k_y a - \cos k_z a)$ where the system is gapless. This happens when the k dependence of $\Delta_{\nu,\mathbf{k}}$ induces a sign change on a connected Fermi surface, and the zeros of $\Delta_{\nu,\mathbf{k}}$ correspond to the nodal lines. Importantly, there are also dispersionless surface Majorana modes on the surfaces, as shown in Fig. 3d. The topological origin of the Majorana modes can be determined by Eq. (2) as well. Therefore, we conclude that the multifermion material $\text{Li}_2\text{Pt}_3\text{B}$ is likely to be a gapless topological superconductor with Majorana surface modes.

4 Fully gapped pairing phase in $\text{Li}_2\text{Pd}_3\text{B}$

For the case of $\text{Li}_2\text{Pd}_3\text{B}$, with the A_1 representation, the superconducting properties of the system depend on η_1 and η_2 of the order parameter $\Delta_{v,\mathbf{k}} = \eta_1 + \eta_2(3 - \cos k_x a - \cos k_y a - \cos k_z a)$. The topological invariant of the DIII class topological superconductor is given by

$$N_{\text{DIII}} = \frac{1}{2} \sum_v \text{sgn}(\Delta_{v,\mathbf{k}_F}) C_v, \quad (3)$$

where $\Delta_{v,\mathbf{k}_F}$ is the pairing on the Fermi surface of the v -th band with Chern number C_v [43]. Interestingly, there is a large parameter regime of $\Delta_{v,\mathbf{k}_F}$, as shown in Fig. 4e, where the pairing of the large Fermi pockets near Γ and the two Fermi pockets enclosing the R point have different signs. In this case, a DIII class topological superconductor is obtained [44]. In the topological regime, the two nearly

degenerate Fermi pockets enclosing the R point have a total Chern number of 4 and the same pairing sign, while the rest of the Fermi surfaces have a total Chern number of -4 which is opposite to the R pocket. As indicated by Eq. (3), the opposite Chern numbers and pairing sign of the two groups of Fermi surfaces will lead to a finite total topological invariant of the DIII class topological phase $N_{\text{DIII}} = 4$. As a result, we expect that four Majorana cones appear on the (001) surface of the material. This is verified numerically by setting $\eta_1/\eta_2 = -5$, and four Majorana cones are found on the (001) surface state. It is interesting to note that on the (111) surface, the C_3 rotational symmetry forces the Majorana cones to be pinned at the surface Brillouin zone center $\bar{\Gamma}$ and results in two Majorana cones with linear and cubic dispersion, respectively, as shown in Fig. 4c and 4d. This cubic dispersion Majorana cone is a manifestation of the existence of multifold fermions, particularly the spin-3/2 fermions, as pointed out in [45, 46].

We would like to emphasize that the topological superconducting phase in $\text{Li}_2\text{Pd}_3\text{B}$ can be achieved without an odd parity pairing, which is sometimes referred to as Fu-Berg criteria [21, 38, 45–48]. Since in previous systems, the opposite Chern numbers originate from different spin channel, which naturally requires a spin-triplet pairing to achieve non-trivial topology given by Eq. (3). In contrast, the different Chern numbers in $\text{Li}_2\text{Pd}_3\text{B}$ are from multifold fermions, which does not necessarily require ASOC.

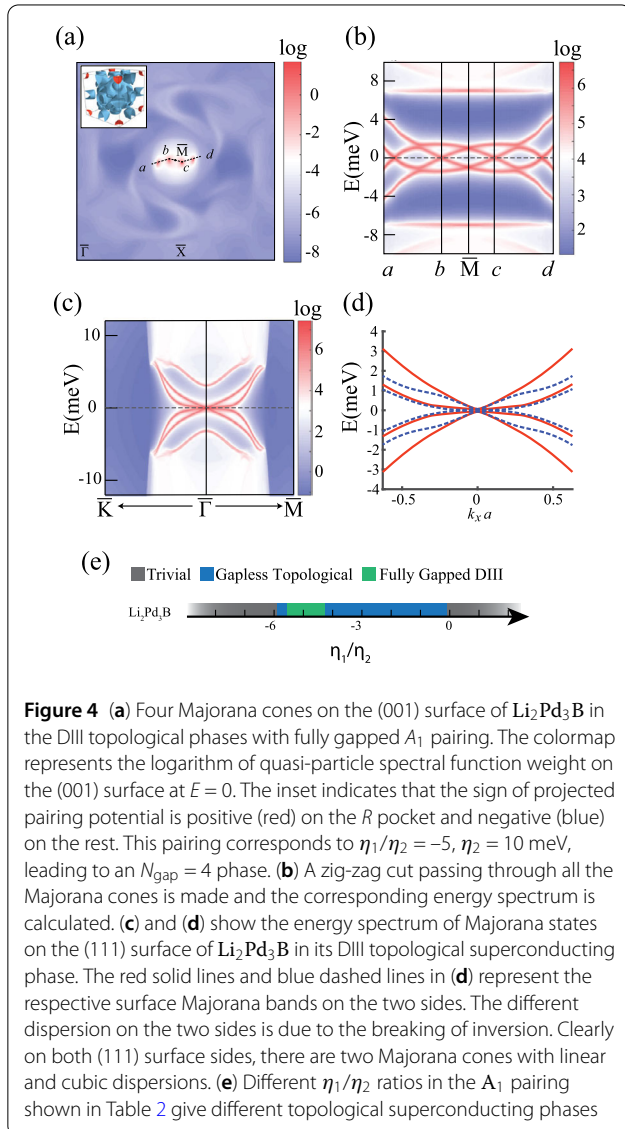
5 Other superconducting materials with multifold fermions

In this work, we show that the normal state topology plays an important role in determining the superconducting properties of $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$. We expect that the normal state topological properties are important in many other noncentrosymmetric superconductors as well. By revisiting the space group of noncentrosymmetric superconductors [21, 49–52], we can further identify several superconductors with multifold fermions, such as $\text{Mo}_3\text{Al}_2\text{C}$, PdBiSe , Y_2C_3 and La_2C_2 whose normal state topological properties have been ignored in previous studies.

Their first-principles band structures are shown in Fig. S2 in Additional file 1. Due to their nonsymmorphic symmetries, plentiful unconventional fermions with 3-, 4-, 6- and 8-fold degeneracy appear at high-symmetry k points, and some of them are near the Fermi level and thus influence the superconducting phases. Their possible topological superconducting phases can be further studied with the same numerical and theoretical methods presented in this work.

6 Conclusion

In this work, we showed that $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$ are multifold fermion metals with unconventional degenerate points at time-reversal invariant momenta due to nonsymmorphic symmetry. This results in Fermi surfaces with



higher Chern numbers, even in the absence of ASOC. We showed that there are long Fermi arc states in $\text{Li}_2\text{Pd}_3\text{B}$ at the Fermi energy. Interestingly, $\text{Li}_2\text{Pd}_3\text{B}$, being a fully gapped superconductor, is a candidate material of a DIII class topological superconductor. On the other hand, superconducting $\text{Li}_2\text{Pt}_3\text{B}$, being gapless, is likely to be a nodal topological superconductor with dispersionless Majorana surface modes. We also identified several other superconducting multifold fermion superconductors, which can also provide new platforms to study the interplay between normal state topology and superconductivity.

Supplementary information

Supplementary information accompanies this paper at <https://doi.org/10.1007/s44214-022-00001-1>.

Additional file 1. Supplementary information (PDF 1.3 MB)

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Availability of data and materials

The data and materials that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests

The authors declare that they have no competing interests.

Author contribution

KTL supervised the project. ZSG and X-JG performed the major part of the calculations and material analysis. X-JG and XYX performed the DFT calculation. KTL, ZSG and X-JG wrote the manuscript with contributions from all authors. W-YH and TKN discussed the results and commented on the manuscript. All authors read and approved the final manuscript.

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Supplementary Material:

Topological Superconductivity from Multi-fold Fermions in Chiral Crystal $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$

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A. Pairing symmetry classification

To find the possible pairing states and analyze their quasi-particle spectrum, one first has to classify all the pairing channels according to the point group symmetry. For multi-dimensional representations, symmetry has to be further identified by finding the local minima (stationary points) of the corresponding Ginzburg-Landau free energy. Resulting symmetry is described by the isotropy group, which can then be used to find the possible quasi-particle gap structures [S1–S3]. For the normal state in the presence of SOC, the full symmetry group should be $G_{total} = G \times U(1) \times \mathcal{T}$, where G is the double space group of the lattice. In the vicinity of T_c , the symmetry group is spontaneously lowered into the subgroup G'_{total} of G_{total} . Explicitly speaking, $g \in G'_{total}$ should be an element of $H \subset G$, accompanied by a phase factor from $U(1)$ or a complex conjugate from $\mathcal{T}$. We list these in table S1 in the form of isotropy group $H(\Gamma_i)$. It can be understood that if a pairing is invariant under $H(\Gamma_i)$, it transforms as if it is the i -th irreducible representation (IR) Γ_i , of H .

The H reflects the symmetry of energetics $E_{\mathbf{k}}$, which can be probed in spectroscopic measurements. We did not consider any further mixing due to the symmetry breaking of the pairing potential [S2].

A₁ pairing— This pairing can be connected to the trivial s-wave pairing. Nodal structure is accidental. In this case, as argued in the main text, it is a possible DIII class topological superconductor, supporting chiral majorana fermions at its surface.

A₂ pairing— In this phase, $C_{2,xy}$ and $C_{4,z}$ are accompanied by an $e^{i\pi}$ phase. This forces nodes in pairing potential, which is discussed in the next section.

E pairing— In the E -pairing case, the homogeneous GL free energy density can be expressed as

$$f = \alpha|\boldsymbol{\eta}|^2 + \beta_1|\boldsymbol{\eta}|^4 + \beta_2(\eta_1^*\eta_2 - \eta_2^*\eta_1)^2 + \gamma_1|\boldsymbol{\eta}|^6 + \gamma_2|\boldsymbol{\eta}|^2|\eta_1^2 + \eta_2^2|^2 + \gamma_3|\eta_1^2||3\eta_2^2 - \eta_1^2|^2 \quad (\text{S1})$$

where $\boldsymbol{\eta} = (\eta_1, \eta_2)$ is the two component order parameter, and α, β, γ are material dependent parameters. The sixth order term has to be included to split accidental degeneracy. Local minima of the free energy can be found at $(1, 0)$, $(0, 1)$ and $(1, i)$. A multi-component order parameter allows a complex configuration $(1, i)$, spontaneously breaking TR symmetry.

T₁/T₂ pairing. — In both cases, the order parameter has three components $\boldsymbol{\eta} = (\eta_1, \eta_2, \eta_3)$, and the GL free energy density can be written as $f = \alpha|\boldsymbol{\eta}|^2 + \beta_1|\boldsymbol{\eta}|^4 + \beta_2|\boldsymbol{\eta} \cdot \boldsymbol{\eta}|^2 + \beta_3(|\eta_1|^2|\eta_2|^2 + |\eta_2|^2|\eta_3|^2 + |\eta_3|^2|\eta_1|^2)$. Now the distinct ground states are: $(1, 0, 0)$, $(1, 1, 1)$, $(1, \omega, \omega^2)$ and $(1, i, 0)$, with $\omega = e^{i2\pi/3}$. Again, the last two terms are related to the time-reversal symmetry breaking (TRSB) phase. According to Table S1, the saddle points of multi-dimensional IRs are either nematic or TRSB, and only $A_{1/2}$ preserves the full point group symmetry and TR.

B. Symmetry of the pairing function

Time-reversal symmetry can be written as $\mathcal{T} = C\mathcal{K}$ with $\mathcal{T}^2 = -1$ (spinful) so that

$$C_{\mathbf{k}}C_{\mathbf{k}}^\dagger = 1 \quad C_{\mathbf{k}} = -C_{-\mathbf{k}}^T, \quad \mathcal{K}a = a^*\mathcal{K}, \quad \text{for } a \in \mathbb{C}. \quad (\text{S2})$$

Then we have:

$$\mathcal{T}H\mathcal{T}^{-1} = H \quad \Rightarrow \quad C_{-\mathbf{k}}H_{-\mathbf{k}}^*C_{-\mathbf{k}}^\dagger = H_{\mathbf{k}}. \quad (\text{S3})$$

Also, TR symmetry has the properties that given a space group symmetry $g \in G$:

$$g\mathcal{T}|\psi_{\mathbf{p},\alpha}\rangle = \mathcal{T}g|\psi_{\mathbf{p},\alpha}\rangle \Rightarrow CD(g)^* = D(g)C. \quad (\text{S4})$$

This indicates that the C matrix can change a representation to its conjugate (adjoint) partner.

IR	Stationary points	Time-Reversal	Isotropy group H	Node type	$\Delta_{\mathbf{k}}$
A_1	–	TR	$O(A_1) : \{C_{3,xyz}, C_{4,z}, C_{2,xy}\}$	–	$1, k_x^2 + k_y^2 + k_z^2$
A_2	–	TR	$O(A_2)\{C_{3,xyz}, -C_{4,z}, -C_{2,xy}\}$	L	$(k_x^2 - k_y^2)(k_y^2 - k_z^2)(k_z^2 - k_x^2)$
E	$(1, 0)$	TR	$D_4(B_1) : \{-C_{4,z}, -C_{2,xy}\}$	L	$\sqrt{3}(k_x^2 - k_y^2)$
	$(0, 1)$	TR	$D_4(A_1) : \{C_{4,z}, C_{2,xy}\}$	–	$2k_z^2 - k_x^2 - k_y^2$
	$(1, i)$	TRSB	$O(E) : \{\omega C_{3,xyz}, \omega^2 C_{4,z}, \omega^2 C_{2,xy}\mathcal{K}\}$	P	$k_x^2 + \omega k_y^2 + \omega^2 k_z^2$
T_1	$(1, 0, 0)$	TR	$D_4(A_2) : \{C_{4,x}, -C_{2,yz}\}$	L	$k_y k_z (k_y^2 - k_z^2)$
	$(1, 1, 1)$	TR	$D_3(A_2) : \{C_{3,xyz}, -C_{2,x\bar{y}}\}$	L	$k_y k_z (k_y^2 - k_z^2) + k_z k_x (k_z^2 - k_x^2) + k_x k_y (k_x^2 - k_y^2)$
	$(1, \omega, \omega^2)$	TRSB	$D_3(E) : \{\omega C_{3,xyz}, -\omega C_{2,x\bar{y}}\mathcal{K}\}$	P	$k_y k_z (k_y^2 - k_z^2) + \omega k_z k_x (k_z^2 - k_x^2) + \omega^2 k_x k_y (k_x^2 - k_y^2)$
	$(1, i, 0)$	TRSB	$D_4(E) : \{iC_{4,z}, -C_{2,y}\mathcal{K}\}$	L	$k_y k_z (k_y^2 - k_z^2) + i k_z k_x (k_z^2 - k_x^2)$
T_2	$(1, 0, 0)$	TR	$D_4(B_2) : \{-C_{4,x}, -C_{2,y}\}$	L	$k_y k_z$
	$(1, 1, 1)$	TR	$D_3(A_1) : \{C_{3,xyz}, C_{2,x\bar{y}}\}$	–	$k_y k_z + k_z k_x + k_x k_y$
	$(1, \omega, \omega^2)$	TRSB	$D_3(E) : \{\omega C_{3,xyz}, \omega C_{2,x\bar{y}}\mathcal{K}\}$	P	$k_y k_z + \omega k_z k_x + \omega^2 k_x k_y$
	$(1, i, 0)$	TRSB	$D_4(E) : \{iC_{4,z}, -C_{2,y}\mathcal{K}\}$	L	$k_y k_z + i k_z k_x$

TABLE S1. The local minima of various pairing states. TR and TRSB indicate time-reversal and time-reversal symmetry breaking phases, respectively. The isotropy group H describes the symmetry of the minima, which is isomorphic to the subgroup of the O . The explicit form of the generators is also given. For the node type in single band pairing, ‘ P ’ denotes symmetry protected point nodes, ‘ L ’ denotes line nodes, and ‘–’ denotes that there is possibility of finding accidental nodes. Examples of basis functions are given in the last column. These may have accidental nodes that will be eliminated when higher-order terms of $\mathbf{k}$ are included. Also note that when projected to a specific band, the pairing function should always be an even function of $\mathbf{k}$. Here $\omega = e^{i2\pi/3}$.

Now given a superconducting BdG Hamiltonian:

$$\mathcal{H}_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} H_{\mathbf{k}} & \Delta_{\mathbf{k}} \\ \Delta_{\mathbf{k}}^\dagger & -H_{-\mathbf{k}}^T \end{pmatrix}, \quad (\text{S5})$$

it is very helpful to define a TR basis when TR symmetry is present:

$$\Psi_{\mathbf{k}} = [\psi_{\mathbf{k}}, \mathcal{T}\psi_{\mathbf{k}}^\dagger \mathcal{T}^{-1}]^T = [\psi_{\mathbf{k}}, C_{\mathbf{k}}^T \psi_{-\mathbf{k}}^\dagger]^T. \quad (\text{S6})$$

In this basis, the BdG Hamiltonian now reads:

$$\begin{aligned} \tilde{\mathcal{H}}_{\text{BdG}}(\mathbf{k}) &= \begin{pmatrix} \mathbb{1} & \\ & C_{\mathbf{k}}^T \end{pmatrix} \begin{pmatrix} H_{\mathbf{k}} & \Delta_{\mathbf{k}} \\ \Delta_{\mathbf{k}}^\dagger & -H_{-\mathbf{k}}^T \end{pmatrix} \begin{pmatrix} \mathbb{1} & \\ & C_{\mathbf{k}}^* \end{pmatrix} \\ &= \begin{pmatrix} H_{\mathbf{k}} & \tilde{\Delta}_{\mathbf{k}} \\ \tilde{\Delta}_{\mathbf{k}} & -H_{\mathbf{k}} \end{pmatrix} = \tau_z \otimes H_{\mathbf{k}} + \tau_x \otimes \tilde{\Delta}_{\mathbf{k}}. \end{aligned} \quad (\text{S7})$$

Note that the last line only holds when TR symmetry is present. Now the new pairing function, $\tilde{\Delta}_{\mathbf{k}} = \Delta_{\mathbf{k}} C_{\mathbf{k}}^*$ has the following properties required by TR/space group symmetry and Fermi statistics:

$$\begin{aligned} \mathcal{T} : \quad & \tilde{\Delta}_{\mathbf{k}}^\dagger = \tilde{\Delta}_{\mathbf{k}}, \\ g : \quad & D(g) \tilde{\Delta}_{\mathbf{k}} D(g)^\dagger = \tilde{\Delta}_{g\mathbf{k}}, \\ \text{Fermi statistics :} \quad & \tilde{\Delta}_{\mathbf{k}} = C_{-\mathbf{k}} \tilde{\Delta}_{-\mathbf{k}}^T C_{-\mathbf{k}}^\dagger \end{aligned} \tag{S8}$$

Then we can expand this function $\tilde{\Delta}_{\mathbf{k}} = \sum_{i,\mu} \eta_{i\mu} \Gamma_{i\mu}(\mathbf{k})$ into IRs of the point group. $\eta_{i\mu}$ is the corresponding order parameter in Ginzburg-Landau (GL) free energy, which is real or complex when TR symmetry is present or absent, respectively.

At general points in chiral superconductors, strong asymmetric spin-orbit coupling lifts all degeneracy and superconductivity pairs electrons within a single band. This is also in line with the superconductivity fitness condition of [S5–S7], which states that pairing is most stable when $H_0(\mathbf{k})$ and $\tilde{\Delta}_0(\mathbf{k})$ are simultaneously diagonalized: $[H_0(\mathbf{k}), \tilde{\Delta}_0(\mathbf{k})] = 0$. In this eigen-band basis we can write the simple Bogoliubov-de Gennes Hamiltonian as

$$\tilde{\mathcal{H}}_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \text{diag}[\xi_{\nu,\mathbf{k}}] & \text{diag}[\Delta_{\nu,\mathbf{k}}] \\ \text{diag}[\Delta_{\nu,\mathbf{k}}^*] & -\text{diag}[\xi_{\nu,\mathbf{k}}] \end{pmatrix}. \tag{S9}$$

TR matrix C is an odd function of $\mathbf{k}$, such that $t(\mathbf{k}) = -t(\mathbf{k})$ and $|t(\mathbf{k})| = 1$. For the pairing function $\tilde{\Delta}(\mathbf{k})$, it is an even function of $\mathbf{k}$. We discuss such pairing in the computations within the main text. For such single band pairing, symmetries with an additional $U(1)$ phase will stabilize nodes on the Fermi surface. For example, in an A_2 phase, there is a π phase accompanied with $C_{4,z}$ rotation. Then, there must be at least two line nodes traversing through the rotation axis, since the pairing potential is a real function of $\mathbf{k}$. For TRSB phase such as the $(1, i)$ for E , the C_3 has a $e^{i2\pi/3}$ phase. However, since the time-reversal symmetry is broken, we can only conclude that the pairing function is zero at the intersection between the rotation axis and Fermi surface, as the phase is not well defined only at this point. We list all the node types in Table S1.

On the Brillouin zone boundary, non-symmorphic symmetry combined with the time-reversal symmetry will force Kramers-type degeneracy. If a line node as classified in the last paragraph resides in this degenerate point, it is then necessary to ask whether this node is susceptible to inter-band pairing. Luckily, according to the results in Table S1, the line nodes in general are not located on the Brillouin zone boundary, thus inter-band pairing does not affect our result.

For the results of the superconducting case, we first adopted the basis defined in Eq. S7 to write the BdG Hamiltonian: $\mathcal{H}_{\mathbf{k}} = H_{\mathbf{k}} \otimes \tau_z + \tilde{\Delta}_{\mathbf{k}} \otimes \tau_x$. For the A_1 case, the pairing takes the form

$$\begin{aligned} \tilde{\Delta}_{\mathbf{k}} = & \{d_0 + d_1 [6 - 2 \cos(k_x) - 2 \cos(k_y) \\ & - 2 \cos(k_z)]\} \mathbb{1}_{8 \times 8}. \end{aligned} \tag{S10}$$

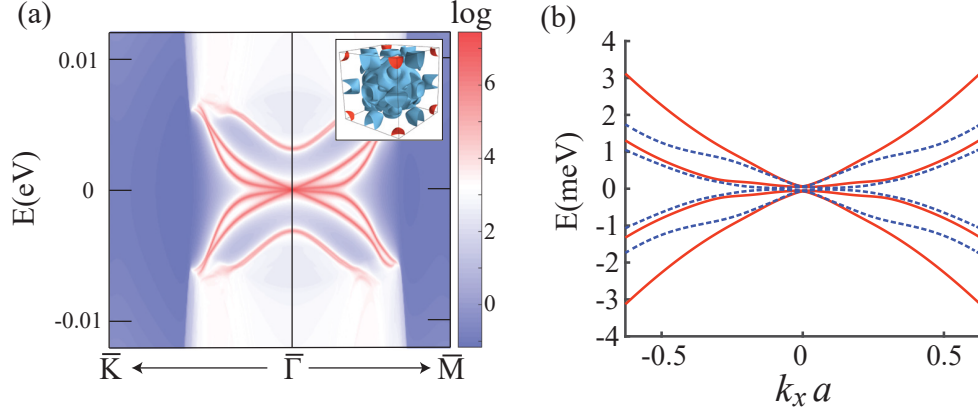


FIG. S1. The spectral function on the (111) left surface of DIII topologically superconducting $\text{Li}_2\text{Pd}_3\text{B}$ is shown in (a), where a linear Majorana cone and a cubic Majorana cone at $\bar{\Gamma}$ can be clearly identified. The inset in (a) indicates that the sign of projected pairing potential is positive (red) on the R pocket and negative (blue) on the rest. In (b), we present the Majorana spectrum of the surface projected 8-by-8 Hamiltonian, and the blue (red) curves represent the bands on the left (right) side surface. The different dispersion of the Majorana bands on the two surfaces is due to the broken inversion symmetry, but it is the same for the two surfaces that upon each of them there is one linear Majorana cone and one cubic Majorana cone.

And the parameters for DIII topological phases ($N = 4$) in Fig. 3 in the main text are $(d_0, d_1) = (-0.05, 0.005)$ (in units of eV). For A_2 pairing, it is taken as

$$\begin{aligned} \tilde{\Delta}_{\mathbf{k}} = & d_0 (\cos k_x a - \cos k_y a) \\ & \times (\cos k_y a - \cos k_z a) (\cos k_z a - \cos k_x a) \mathbb{1}_{8 \times 8}, \end{aligned} \quad (\text{S11})$$

where the size of d_0 doesn't affect the physics and is set to be 0.02 eV.

C. Cubic dispersion Majorana bands on the (111) surface of DIII topological superconducting $\text{Li}_2\text{Pd}_3\text{B}$

As described in the main text, for the case of $\text{Li}_2\text{Pd}_3\text{B}$ with the A_1 representation, the superconducting properties of the system depend on η_1 and η_2 of the order parameter $\Delta_{\nu, \mathbf{k}} = \eta_1 + \eta_2 (3 - \cos k_x a - \cos k_y a - \cos k_z a)$. We shall here study the dispersion of Majorana fermions on the (111) surface under symmetry constraint. Numerically we first set $\eta_1/\eta_2 = -5$, and obtain the eight eigen-states nearest to the zero energy at $\bar{\Gamma}$ and projected the complicated slab Wannier Hamiltonian to them to get the 8-by-8 surface Hamiltonian. The energy spectrum of this projected surface Hamiltonian is shown in Fig. 3d in main text, and the linear as well as the cubic dispersion were verified by suitable fitting. It should be noted that the band difference between the two surfaces is due to the lack of inversion symmetry.

It is pointed out in [S10] that the cubic Majorana cone can result from spin-3/2 fermions and is protected by time-reversal $\hat{\mathcal{T}}$, particle-hole $\hat{\mathcal{P}}$ and $\hat{C}_3$ rotation symmetry, which is exactly the case for the (111) surface of $\text{Li}_2\text{Pd}_3\text{B}$. The argument is as follows. Assuming that the Majorana surface states are formed from spin-3/2 fermions, the symmetry operators mentioned above and their restrictions on the surface Hamiltonian can be written as

$$\begin{aligned}\hat{\mathcal{T}} &= i\sigma_y\hat{\mathcal{K}}, \quad \hat{\mathcal{T}}^{-1}H(\mathbf{k})\hat{\mathcal{T}} = H(-\mathbf{k}), \\ \hat{\mathcal{P}} &= \sigma_x\hat{\mathcal{K}}, \quad \hat{\mathcal{P}}^{-1}H(\mathbf{k})\hat{\mathcal{P}} = -H(-\mathbf{k}), \\ \hat{C}_3 &= -\sigma_0, \quad \hat{C}_3^{-1}H(\hat{C}_3\mathbf{k})\hat{C}_3 = H(\mathbf{k}).\end{aligned}\tag{S12}$$

Simple calculation tells us that $\hat{\mathcal{T}}$ together with $\hat{\mathcal{P}}$ requires that the surface Hamiltonian should only contain $k^{(odd)}\sigma_{x,y}$ terms while $\hat{C}_3$ requires that $H(k_+, k_-) = H(k_+e^{i\frac{2}{3}\pi}, k_-e^{-i\frac{2}{3}\pi})$ with $k_{\pm} = k_x \pm ik_y$. Thus, to the lowest-allowed k terms, the surface Majorana Hamiltonian is

$$H(k_+, k_-) = c_1(k_+^3\sigma_+ + k_-^3\sigma_-) + c_2(k_+^3\sigma_- + k_-^3\sigma_+),\tag{S13}$$

where $\sigma_{\pm} = \sigma_x \pm i\sigma_y$. In the contrast, for spin-1/2 fermions, the lowest-allowed terms are linear in k . In the weak pairing limit, the surface Majorana states are just linear combinations of particle and hole states near the Fermi level of bulk $\text{Li}_2\text{Pd}_3\text{B}$. In this sense, the existence of cubic Majorana bands is the direct result of the spin-3/2 unconventional fermions on the Fermi surfaces of $\text{Li}_2\text{Pd}_3\text{B}$.

D. Topological Invariants of the Gapless Superconductor $\text{Li}_2\text{Pt}_3\text{B}$

Throughout this work, we study the superconducting system with particle-hole symmetry $\hat{\mathcal{P}}$, time-reversal symmetry $\hat{\mathcal{T}}$ and their combination the chiral symmetry $\hat{\mathcal{C}} = i\hat{\mathcal{P}}\hat{\mathcal{T}}$, which makes the fully-gapped superconducting $\text{Li}_2\text{Pd}_3\text{B}$ classified into the 3-d DIII topological class with a $\mathbb{Z}$ topological invariant [S8]. In comparison, for the gapless superconducting $\text{Li}_2\text{Pt}_3\text{B}$ with nodal lines, to define an overall topological invariant upon the 3-d first BZ is not possible. However, it is still possible to define a $\mathbb{Z}$ topological invariant for any 1-d loop that do not intersect with any nodal line, because with the preserved chiral symmetry $\hat{\mathcal{C}}$ this loop the system can be classified into the 1-d AIII topological class. We have provided a way to define this $\mathbb{Z}$ topological invariant in Eq.2 in the main text. Here, we further justify it.

In Sec. B, we have written the BdG Hamiltonian in the TR basis as

$$\begin{aligned}
\tilde{\mathcal{H}}_{\text{BdG}}(\mathbf{k}) &= V \mathcal{H}_{\text{BdG}}(\mathbf{k}) V^\dagger \\
&= \begin{pmatrix} \mathbb{1} & \\ & C_{\mathbf{k}}^T \end{pmatrix} \begin{pmatrix} H_{\mathbf{k}} & \Delta_{\mathbf{k}} \\ \Delta_{\mathbf{k}}^\dagger & -H_{\mathbf{k}}^T \end{pmatrix} \begin{pmatrix} \mathbb{1} & \\ & C_{\mathbf{k}}^* \end{pmatrix} . \\
&= \begin{pmatrix} H_{\mathbf{k}} & \tilde{\Delta}_{\mathbf{k}} \\ \tilde{\Delta}_{\mathbf{k}} & -H_{\mathbf{k}} \end{pmatrix} = \tau_z \otimes H_{\mathbf{k}} + \tau_x \otimes \tilde{\Delta}_{\mathbf{k}} .
\end{aligned} \tag{S14}$$

Under this basis, the chiral symmetry can be simply formulated as

$$\tilde{U}_C = \tau_y, \tag{S15}$$

with $(\tilde{U}_C)^2 = \mathbb{1}$ and $\{\tilde{\mathcal{H}}_{\text{BdG}}(\mathbf{k}), \tilde{U}_C\} = 0$. We further rewrite the BdG Hamiltonian in the eigen-basis of the chiral symmetry as

$$\begin{aligned}
\tilde{\mathcal{H}}'_{\text{BdG}}(\mathbf{k}) &= V' \tilde{\mathcal{H}}_{\text{BdG}}(\mathbf{k}) V'^\dagger \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbb{1} & -i \\ & i \end{pmatrix} \begin{pmatrix} H_{\mathbf{k}} & \tilde{\Delta}_{\mathbf{k}} \\ \tilde{\Delta}_{\mathbf{k}} & -H_{\mathbf{k}} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbb{1} & \mathbb{1} \\ i & -i \end{pmatrix} \\
&= \begin{pmatrix} & Q_{\mathbf{k}} \\ Q_{\mathbf{k}}^\dagger & \end{pmatrix}
\end{aligned} \tag{S16}$$

with $Q_{\mathbf{k}} = H_{\mathbf{k}} - i\tilde{\Delta}_{\mathbf{k}}$.

With these preliminary matrix transformations, we now justify that the quantity

$$N_C = \frac{1}{2\pi} \text{Im} \int_C d\mathbf{k} \text{Tr}[\partial_{\mathbf{k}} \ln Q_{\mathbf{k}}] \tag{S17}$$

is indeed a quantized topological invariant. Here $\mathcal{C}$ is a closed loop that do not intersect with any nodes. As we are only concerned about the weak pairing limit, $H_0(\mathbf{k})$ and $\tilde{\Delta}_0(\mathbf{k})$ can be simultaneously diagonalized as

$$H_{\mathbf{k}} = \text{diag}[\xi_{\nu,\mathbf{k}}], \quad \tilde{\Delta}_{\mathbf{k}} = \text{diag}[\Delta_{\nu,\mathbf{k}}], \tag{S18}$$

with real numbers $\xi_{\nu,\mathbf{k}}$ and $\Delta_{\nu,\mathbf{k}}$ for each band ν . Then we have

$$N_{\text{AIII},C} = \frac{1}{2\pi} \int_C d\mathbf{k} \partial_{\mathbf{k}} [\text{Im} \text{Tr} \ln Q_{\mathbf{k}}] = \frac{1}{2\pi} \sum_{\nu} \int_C d\mathbf{k} \partial_{\mathbf{k}} [\arg(q_{\nu,\mathbf{k}})] \tag{S19}$$

with $q_{\nu,\mathbf{k}} = \xi_{\nu,\mathbf{k}} - i\Delta_{\nu,\mathbf{k}}$. Now we can easily see that $N_{\text{AIII},C}$ is the summation of the winding numbers of $q_{\nu,\mathbf{k}}$ around the origin on the complex plane, which is a $\mathbb{Z}$ topological invariant.

Moreover, when studying the surface states, this 1-d loop can be chosen to be any straight line, which goes across the 1st BZ along the normal direction of the surface and also do not intersect with any nodal lines. Now the topological invariant $N_{\text{AIII},C}$

we just defined has a physical meaning – the number of Majorana zero modes at each end of this quasi-1d system. At one end, all the Majorana zero modes have the same chiral eigen-value C_n and this condition protects them from coupling to each other and being lifted to a finite energy as

$$\begin{aligned}\langle \psi_n | \hat{\mathcal{H}}_{BdG} | \psi_{n'} \rangle &= \langle \psi_n | \hat{C} \hat{C} \hat{\mathcal{H}}_{BdG} | \psi_{n'} \rangle \\ &= -\langle \psi_n | \hat{C} \hat{\mathcal{H}}_{BdG} \hat{C} | \psi_{n'} \rangle \\ &= -C_n C_{n'} \langle \psi_n | \hat{\mathcal{H}}_{BdG} | \psi_{n'} \rangle,\end{aligned}\tag{S20}$$

$$(1 + C_n C_{n'}) \langle \psi_n | \hat{\mathcal{H}}_{BdG} | \psi_{n'} \rangle = 0.\tag{S21}$$

In short, $\langle \psi_n | \hat{\mathcal{H}}_{BdG} | \psi_{n'} \rangle$ is forced to be zero when $C_n C_{n'} = 1$.

E. $\mathbf{k} \cdot \mathbf{p}$ model and explicit matrix representations of the little groups

To analyze the low energy effect model of multi-fold fermions, it is ideal to construct a $\mathbf{k} \cdot \mathbf{p}$ model. We focus on the linear terms of $\mathbf{k}$, so that the Hamiltonian can be written in the form:

$$H(\delta \mathbf{k}) = \sum_{i=1,2,3} M^i \delta \mathbf{k}_i,\tag{S22}$$

with $\delta \mathbf{k} = \mathbf{k} - \mathbf{K}_{\Gamma/R}$. The induced representation would be $(\bar{E}_1 \uparrow G) \downarrow G_\Gamma \approx \bar{\Gamma}_6(2) \oplus \bar{\Gamma}_7(2) \oplus \bar{\Gamma}_8(4)$ at the Γ point and $(\bar{E}_1 \uparrow G) \downarrow G_R \approx \bar{R}_6(2) \oplus \bar{R}_7 \bar{R}_8(6)$ at the R point, which can be found at *Bilbao Crystallographic server* [S15]. The numbers in the parentheses indicate the numbers of degeneracy, and $\bar{R}_7$ and $\bar{R}_8$ are related by TR symmetry. Here, $G_{\Gamma/R}$ denotes the little group of point Γ/R . This means that the four-fold fermion is subject to $\bar{\Gamma}_8(4)$, and the two three-fold fermions are subject to $\bar{R}_7(3)$, $\bar{R}_8(3)$, respectively. Then, to find the $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian, for example, of the four-fold fermion at Γ , one just needs to find one set of matrix representations $D(g)$ of $\bar{\Gamma}_8(4)$ for $g \in G_\Gamma$ explicitly and then construct a 4-band $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian $H_{\mathbf{k}}$ that is invariant, such that $D(g)H_{g^{-1}\mathbf{k}}D(g)^\dagger = H_{\mathbf{k}}$. Then, all the allowed terms can be derived from this relation.

1. Four-fold fermion at Γ

For $\bar{\Gamma}_8$ of G_Γ , we use the following form [S15]:

$$\begin{aligned}\{1|0\} &= \mathbb{1}_{4 \times 4}, & \{2_{001}|\frac{1}{2}0\frac{1}{2}\} &= \begin{pmatrix} -i & & & \\ & i & & \\ & & i & \\ & & & -i \end{pmatrix}, & \{2_{010}|0\frac{1}{2}\frac{1}{2}\} &= \begin{pmatrix} & -1 & & \\ 1 & & & \\ & & -i & \\ & & & -i \end{pmatrix}, \\ \{1^d|0\} &= -\mathbb{1}_{4 \times 4}, & \{3_{111}^+|0\} &= \frac{\sqrt{2}}{2} \begin{pmatrix} e^{i5\pi/12} & e^{-i\pi/12} & & \\ e^{i5\pi/12} & e^{i11\pi/12} & & \\ & e^{-i5\pi/12} & e^{i7\pi/12} & \\ & e^{-i11\pi/12} & e^{-i11\pi/12} & \end{pmatrix}, & \{2_{110}|\frac{1}{4}\frac{3}{4}\frac{1}{4}\} &= \begin{pmatrix} & -1 & & \\ 1 & & & \\ & & 1 & \\ & & & -1 \end{pmatrix}.\end{aligned}\tag{S23}$$

Here $\{1^d|0\}$ denotes the 2π rotation. We first get two sets of matrices so that $M^i = v_1 F_1^i + v_2 F_2^i$:

$$\begin{aligned} F_1^1 &= \begin{pmatrix} \sigma_x & \\ & \sigma_y \end{pmatrix}, F_1^2 = \begin{pmatrix} \sigma_y & \\ & \sigma_x \end{pmatrix}, F_1^3 = \begin{pmatrix} \sigma_z & \\ & -\sigma_z \end{pmatrix}, \\ F_2^1 &= \begin{pmatrix} 0 & 0 & e^{i\pi/12} & 0 \\ 0 & 0 & 0 & -ie^{i\pi/12} \\ e^{-i\pi/12} & 0 & 0 & 0 \\ 0 & ie^{-i\pi/12} & 0 & 0 \end{pmatrix}, F_2^2 = \begin{pmatrix} 0 & 0 & -e^{-i\pi/12} & 0 \\ 0 & 0 & 0 & -ie^{-i\pi/12} \\ -e^{i\pi/12} & 0 & 0 & 0 \\ 0 & ie^{-i\pi/12} & 0 & 0 \end{pmatrix}, \\ F_2^3 &= \begin{pmatrix} 0 & 0 & 0 & e^{i\pi/4} \\ 0 & 0 & e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} & 0 & 0 \\ e^{-i\pi/4} & 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (\text{S24})$$

The four-fold fermion at the Γ point of a cubic lattice was previously identified as a spin-3/2 fermion [S1, S17–S21]. To better connect this effective Hamiltonian to the spin-3/2 fermion, we recombine the two sets of F matrices to find the required form: $J^i = -\frac{1}{2}F_1^i + F_2^i$, and $T^i = F_1^i + \frac{1}{2}F_2^i$, so that $H = \mathbf{k} \cdot (v_1' \mathbf{J} + v_2' \mathbf{T})$. In this way, the J^i matrices form a 4-dimensional, spin-3/2 representation of $SO(3)$: $[J^i, J^j] = i\epsilon^{ijk} J^k$ and $\sum J^2 = \sqrt{\frac{1}{2}l(l+1)}\mathbb{1}$ with $l = 3/2$. And the remaining T matrices introduce the effect of a cubic crystalline field.

2. Three-fold fermion at R

At the R point, the 6-fold degeneracy is composed of an $\bar{R}_7 \bar{R}_8$ representation, where $\bar{R}_7$ and $\bar{R}_8$ are connected by TR symmetry. They are interpreted as two spin-1 fermions. We can first write one of the spin-1 fermion as $H_{\mathbf{k}} = \mathbf{k} \cdot \Gamma$. We project the Γ matrices to the $\bar{R}_7$ representation, whose explicit form is

$$\begin{aligned} \{1|0\} &= \mathbb{1}_{3 \times 3}, & \{2_{001}|\frac{1}{2}0\frac{1}{2}\} &= \begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \end{pmatrix}, & \{2_{010}|0\frac{1}{2}\frac{1}{2}\} &= \begin{pmatrix} 1 & & \\ & -1 & \\ & & 1 \end{pmatrix}, \\ \{1^d|0\} &= -\mathbb{1}_{3 \times 3}, & \{3_{111}^+|0\} &= \begin{pmatrix} & & \\ 1 & & \\ & & -1 \end{pmatrix}, & \{2_{110}|\frac{1}{4}\frac{3}{4}\frac{1}{4}\} &= \begin{pmatrix} & & \\ & -i & \\ -i & & \end{pmatrix}. \end{aligned} \quad (\text{S25})$$

The same can be done for $\bar{R}_8$, whose matrices are the complex conjugate of those of $\bar{R}_7$ since $\bar{R}_8 = \bar{R}_7^*$. The full Hamiltonian can be written as:

$$H_{\delta\mathbf{k}} = \begin{pmatrix} a\delta\mathbf{k} \cdot \Gamma & be^{i\theta}\delta\mathbf{k} \cdot \Omega \\ be^{-i\theta}\delta\mathbf{k} \cdot \Omega & a\delta\mathbf{k} \cdot \Gamma \end{pmatrix}, \quad (\text{S26})$$

with a, b, θ being real parameters. The resulting Γ^i and Ω^i matrices are:

$$\begin{aligned}\Gamma^1 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \Gamma^2 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \Gamma^3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \Omega^1 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \Omega^2 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \Omega^3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.\end{aligned}\tag{S27}$$

TR operator here reads $\mathcal{T} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathcal{K}$, with $\mathcal{K}$ the conjugate operator. Here, we still have $\mathcal{T}^2 = -1$, despite it being a ‘spin-1’ fermion, as the fundamental block is a spin-1/2 electron. Again, to make the feature of the spin-1 fermion more explicit, we have

$$J^i = \begin{pmatrix} \Gamma^i & \\ & \Gamma^i \end{pmatrix}, \quad T^i = \begin{pmatrix} & e^{i\theta} \Omega^i \\ e^{-i\theta} \Omega^i & \end{pmatrix}.\tag{S28}$$

Here $\sum J^2 = 4\mathbb{1}$, as it represents two copies of a spin-1 fermion. These double spin-1 fermions are also found in previous literature about chiral cubic crystals [S17–S19].

F. Identification of unconventional fermions

The origin of these various types of fermions in lattice systems is from particle physics. Given a relativistic fermion with spin- N , we have a Hamiltonian $H_{\mathbf{k}} = v(\mathbf{k} \cdot \mathbf{J})_{(2N+1) \times (2N+1)}$, with a real coefficient v , and $\mathbf{J} = \{J_x, J_y, J_z\}$ is the $(2N+1)$ dimensional (or spin- N) IR of rotation group $SO(3)$. The energies of the bands are $E_{\mathbf{k}} = vnk$, with $n = -N, -N+1, \dots, N-1, N$. For the n -th band $|n_{\mathbf{k}}\rangle$, the berry curvature can be calculated as

$$\begin{aligned}\mathcal{F}^n(\mathbf{k}) &= -i\epsilon^{ijk} \sum_{m \neq n} \left[\langle \partial_j n_{\mathbf{k}} | m \rangle \langle m | \partial_j n_{\mathbf{k}} \rangle \right] \\ &= -i\epsilon^{ijk} \sum_{m \neq n} \left[\frac{\langle n | \partial_i H | m \rangle \langle m | \partial_j H | n \rangle}{(E_n - E_m)^2} \right],\end{aligned}\tag{S29}$$

whose result can be found by

$$\mathcal{F}^n(\mathbf{k}) = n \frac{\hat{k}}{k^2}.\tag{S30}$$

Then, after integration over the enclosing surface, the Chern number of the n -th band can be found to be $C_n = 2n$, and the monopole charge can be identified as

$$\begin{aligned}
 C_{\text{monopole}} &= \sum_{E_n > 0} C_n \\
 &= \begin{cases} \text{sgn}(v)(N^2 + N) & (\text{integer spin}) \\ \text{sgn}(v)(N + 1/2)^2 & (\text{half integer spin}) \end{cases} \quad (\text{S31})
 \end{aligned}$$

Due to the crystalline symmetry in these solids, the dispersions of these fermions are distorted. However, the Chern numbers of a given band and degeneracy at high symmetry points are unaffected and can be used to identify such fermions. For example, for the 4-fold fermion No. 14 of Table I in the main text, we numerically computed the Chern numbers of the four bands at a tiny sphere around Γ , and we found them to be 3, 1, -1 and -3 (from lower to the higher band). This indicates that the fermion has spin-3/2 with monopole charge -4 . Numerical computation was done with the *WannierTools* package [S27].

G. First-principles calculation of band structures and surface states for $\text{Li}_2\text{Pd}_3\text{B}$ and $\text{Li}_2\text{Pt}_3\text{B}$

All the density functional theory (DFT) [S22] calculations were performed with the *Vienna Ab initio Simulation Package* (VASP) [S23] with the projector-augmented wave method [S24] and the exchange-correlation functional of the Perdew-Berke-Ernzerhof (PBE) form [S25] in the generalized-gradient approximation [S26]. Before we calculated the band structures, the atomic structures of Li_2BPd_3 and Li_2BPt_3 were fully relaxed with the maximum residual force $10^{-4} \text{ eV}\cdot\text{\AA}^{-1}$. The Monkhorst-Pack k-point sampling of $9 \times 9 \times 9$ was adopted for the Brillouin zone integration in the relaxation as well as for the self-consistent calculation. The Wannier tight-binding Hamiltonian was obtained with the *Wannier90* package by projecting the DFT wave functions to the d orbital of Pd(Pt) atoms. We then developed a novel method to get the BdG Hamiltonian by adding the pairing term directly to the Wannier tight binding model under the band basis. For the sake of simplicity, we set the pairing term in the same form for all bands, however, our method could be used in more generalized pairing configurations. With such obtained Wannier tight-binding and BdG Hamiltonians, we were then able to calculate surface spectral functions and topological invariants with the *WannierTools* package [S27].

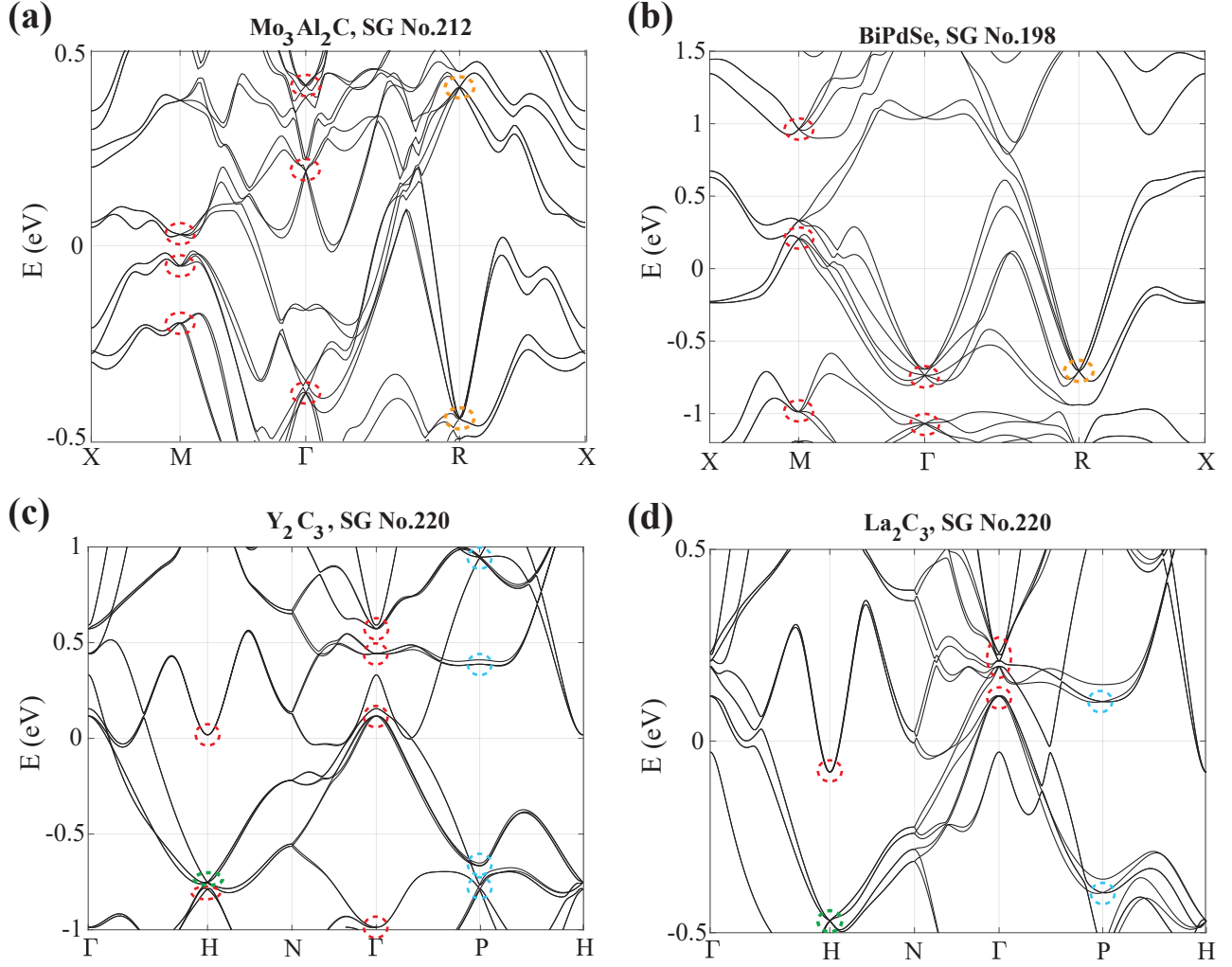


FIG. S2. The first-principles band structure of other multifold fermion superconductors, $\text{Mo}_3\text{Al}_2\text{C}$, BiPdSe , Y_2C_3 and La_2C_3 . The blue, red, orange and green dashed circles highlight the 3-, 4-, 6- and 8-fold fermions, respectively.

H. The first-principles band structure of other superconducting materials with multifold fermions

As mentioned in the main text, we calculated the band structures of superconducting $\text{Mo}_3\text{Al}_2\text{C}$, BiPdSe , Y_2C_3 and La_2C_3 . The results are shown in Fig. S2.

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